TOPIC 2 Systems of Linear Equations and Inequalities

Name _____ Date _____

Reminders
Linear combinations: make one variable's coefficients opposites, then add the equations.
Multiply every term, including the number on the right side.
Fractions: multiply by the LCD first. Variable alone? Substitution is faster.

1. Solve: $2x + y = 11$ and $x - y = 1$.

Tip: add the equations

A. (3, 4)

B. (5, 1)

C. (4, 3)

2. Multiply $\frac{1}{2}x + \frac{3}{4}y = 10$ by 4.

Tip: multiply every term by the LCD

F. $2x + 3y = 40$

G. $2x + 3y = 10$

H. $\frac{1}{2}x + 3y = 40$

3. Which method is fastest for $y = 3x - 1$ and $5x - 2y = 4$?

> **Tip:** variable alone → substitution

A. linear combinations

B. substitution

C. graphing

4. Is (3, 2) a solution of $x + 2y = 7$ and $3x - y = 7$?

> **Tip:** check both equations

F. Only the first is true

G. Only the second is true

H. Yes, both are true